Individual Assignment 5

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Set up

Packages

```
#general use
library(here)

library(tidyverse)

library(janitor)

library(snakecase)

# packages for visualization customization

library(scales)
library(paletteer)
library(ggthemes)

# reading in .xlsx files
library(readxl)

library(NatParksPalettes)

# arranging plots
library(patchwork)
```

Data

```
# taxonomic information
taxon_list <- read_csv(here("data", "taxon_list.csv"))

# aquatic invertebrates
aq_ins <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
                    sheet = "Aquatic Insects")

# site information
sites <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
                   sheet = "sites")

# water quality information
water_quality <- read_xlsx(here("data", "Aquatic Sampling
↪  Data-2026-03-10.xlsx"),
                           sheet = "Water Quality",
                           na = c("", "n/a", "N/A", "over", "173+"))
```

copied from class

```
# creating new object from sites
ncos_sites <- sites |>

  # filter to only include NCOS quarterly sampling sites
  filter(sector == "NCOS" & sampling_frequency == "Quarterly")
```

```
water_quality_clean <- water_quality |>

  # cleaning column names
  clean_names() |>

  # filtering join: only include sites in ncos_sites
  # keep all the rows on the left that match up with rows on the right
  semi_join(ncos_sites, by = "site") |>

  # combine date and site together and remove those indiv. columns
  unite("date_site", date, site, remove = TRUE) |>

  # group by date_site
  group_by(date_site) |>
```

```
  # create new columns that show median of each variable at each date_site
  # because there are multiple samples at the same site at different depths
  summarize(med_ph = median(p_h, na.rm = TRUE),
            med_sal = median(salinity_ppt, na.rm = TRUE),
            med_do = median(dissolved_oxygen_mg_l, na.rm = TRUE)) |>

  # ungroup
  ungroup() |>

  # filter out the salinity outlier
  filter(date_site != "2022-08-12_NVBR")
```

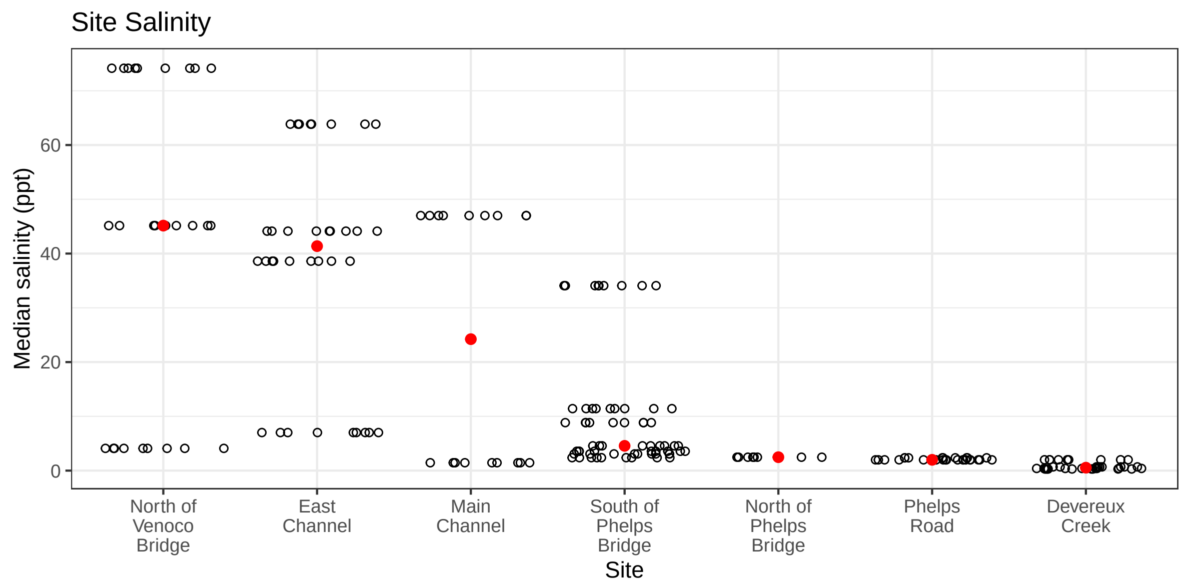
```
# creating new object from taxon_list
taxon_list_clean <- taxon_list |>

  # converted taxon name values to snake case (lowercase)
  mutate(verbatim_name = to_any_case(verbatim_name,
                                     case = "snake"))
```

```
# creating new clean aquatic inverts object from aquatic inverts
aq_ins_clean <- aq_ins |>

  # cleaning column names
  clean_names() |>

  # filtering join: only include quarterly NCOS surveys from ncos_sites
  semi_join(ncos_sites, by = "site") |>

  # renaming column to just date
  rename(date = date_on_vial) |>

  # select columns of interest
  select(date, sample_type, site,

         # 9 most abundant taxon
         ostracod,
         copepod,
         hemiptera_corixidae_boatman,
         cladocera,
         nematode,
         diptera_ceratopogonidae,
```

```
       annelida_oligochaete,
       diptera_chironomid,
       annelida_polychaete) |>

# filter to only include filtered beaker samples
filter(sample_type %in% c("FB250", "FB 250"))  |>


pivot_longer(
  # cols = columns to make longer
  cols = ostracod:annelida_polychaete,
  # new column for the old column names
  names_to = "taxon",
  # new column for the values in each of the old columns
  values_to = "abundance"
) |>

# group by the date, site, and taxon
group_by(date, site, taxon) |>

# calculate average abundance per liter (sum abundance divided by 7.5
 ↪  liters)
summarize(avg_per_lit = sum(abundance, na.rm = TRUE)/7.5) |>

# ungroup
ungroup() |>

 # create a new column called `date_site` to join with water quality
 # don't remove the original columns
unite("date_site", date, site, remove = FALSE) |>

# join with water quality clean
# wqc has `date_site` column as well
left_join(water_quality_clean, by = "date_site") |>

# join with taxonomic information from taxon list
# verbatim_name column in taxon_list_clean matches with taxon column in
 ↪  aq_ins_clean
left_join(taxon_list_clean, by = c("taxon" = "verbatim_name")) |>

# fill in full site names
mutate(site_full = case_when(
  site == "NVBR" ~ "North of Venoco Bridge",
```

```
    site == "NEC" ~ "East Channel",
    site == "NMC" ~ "Main Channel",
    site == "NPB" ~ "South of Phelps Bridge",
    site == "NPB1" ~ "North of Phelps Bridge",
    site == "NPB2" ~ "Phelps Road",
    site == "NWP" ~  "West Pond",
    site == "NDC" ~ "Devereux Creek"
  )) |>

  # setting factor levels for full site name by median salinity
  mutate(site_full = fct_reorder(site_full, med_sal, .fun = "median", .na_rm
   ↪  = TRUE),
         site_full = fct_rev(site_full))
```

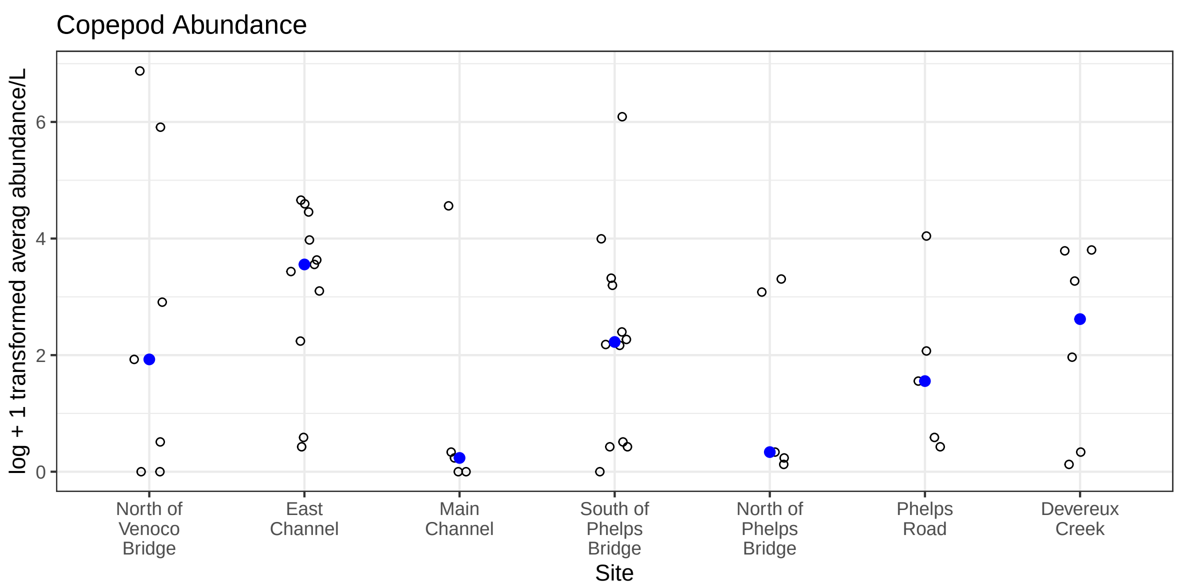
```
salinity_plot <- ggplot(data = aq_ins_clean,
                        # x-axis: full site name
                        # y-axis: median salinity
                        mapping = aes(x = site_full,
                                      y = med_sal)) +
  # points representing median salinity at each survey
  geom_jitter(height = 0,
              shape = 21) +

  stat_summary(geom = "point",
               fun = median,
               color = "red",
               size = 2,) +

  # wrap x-axis labels
  scale_x_discrete(labels = label_wrap(width = 10)) +

  # labeling titles and axes
  labs(x = "Site",
       y = "Median salinity (ppt)",
       title = "Site Salinity") +

  # change theme
  theme_bw()


# displaying the object
salinity_plot
```




```
copepods <- aq_ins_clean |>

  filter(taxon == "copepod") |>

  mutate(log_avg_per_lit = log(avg_per_lit + 1))
```

```
# creating object to store plot
copepod_plot <- ggplot(data = copepods,
                       # x-axis: full site name
                       # y-axis: log transformed avg count/L
                       mapping = aes(x = site_full,
                                     y = log_avg_per_lit)) +
  # jitter to represent salinity
  geom_jitter(height = 0,
              width = 0.1,
              shape = 21) +
# points to represent median copepod abundance
  stat_summary(geom = "point",
               fun = median,
               color = "blue",
               size = 2) +
  # wrapping x axis labels
scale_x_discrete(labels = label_wrap(width = 10)) +

  # relabelling axes
```

```
  labs(x = "Site",
       y = "log + 1 transformed averag abundance/L",
       title = "Copepod Abundance") +

  # different theme
  theme_bw()

# display object
copepod_plot
```



1. Visualizing differences across sites in major taxon abundance

a. Plan your figure

- x-axis: site_full
- y-axis: average abundance of ostracod per liter
- geometry to represent average abundance/L: `geom_jitter()`
- geometry to represent medians: `stat_summary()`

b. Wrangle your data

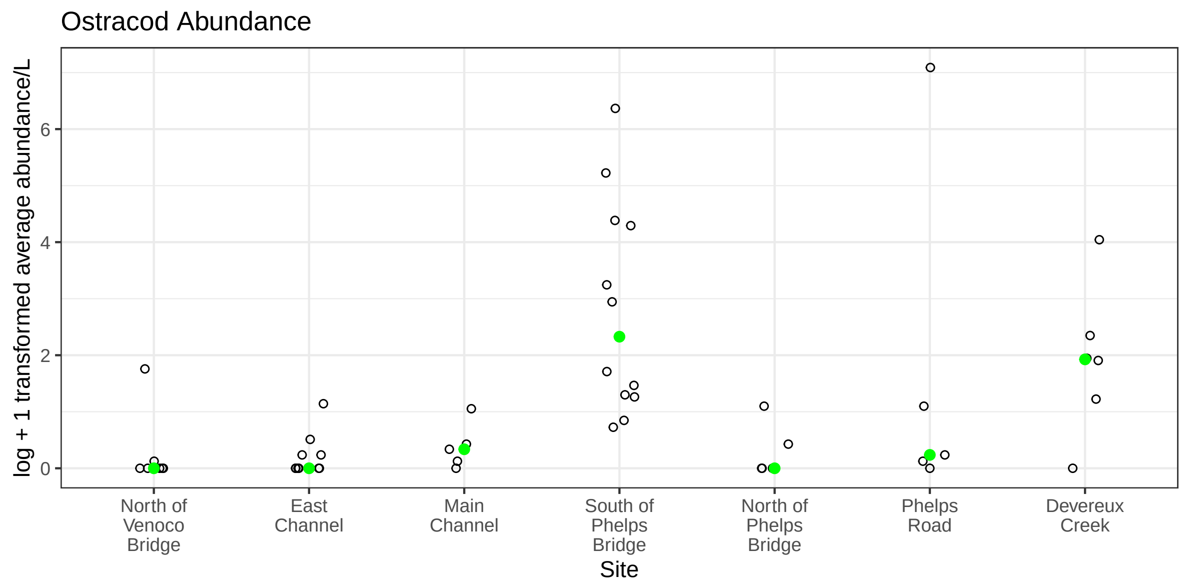
```
# creating new object from aquatic insects
ostracod <- aq_ins_clean |>

  # filter to only include ostracod
  filter(taxon == "ostracod") |>

  # calculate the average of ostracod abundance per lit
  mutate(ostracod_log_avg_per_lit = log(avg_per_lit + 1))

# display 10 random rows

slice_sample(ostracod, n = 10)
```

```
# A tibble: 10 x 24
   date_site   date                site  taxon avg_per_lit med_ph med_sal med_do
   <chr>       <dttm>              <chr> <chr>       <dbl>  <dbl>   <dbl>  <dbl>
 1 2022-02-23~ 2022-02-23 00:00:00 NPB1  ostr~       0      NA      NA     NA
 2 2023-02-28~ 2023-02-28 00:00:00 NPB   ostr~       2.53   NA      NA     NA
 3 2022-11-19~ 2022-11-19 00:00:00 NDC   ostr~       6      NA      NA     NA
 4 2022-11-12~ 2022-11-12 00:00:00 NEC   ostr~       0       7.33    7.01   2.4
 5 2022-02-25~ 2022-02-25 00:00:00 NEC   ostr~       0       8.75   44.1    9.87
 6 2024-01-28~ 2024-01-28 00:00:00 NPB   ostr~       3.33    7.43    3.08   6.6
 7 2023-08-12~ 2023-08-12 00:00:00 NEC   ostr~       0      NA      NA     NA
 8 2023-05-06~ 2023-05-06 00:00:00 NDC   ostr~      56      10.1     NA     13.1
 9 2022-08-15~ 2022-08-15 00:00:00 NMC   ostr~       0.133   7.61   NA     NA
10 2022-05-12~ 2022-05-12 00:00:00 NPB   ostr~       1.07    7.97   11.4    8.17
# i 16 more variables: taxonRank <chr>, scientificName <chr>, kingdom <chr>,
#   Phylum <chr>, Class <chr>, Subclass <chr>, Superorder <chr>, Order <chr>,
#   Infraorder <chr>, Family <chr>, Subfamily <chr>, Tribe <chr>, Genus <chr>,
#   comments <chr>, site_full <fct>, ostracod_log_avg_per_lit <dbl>
```

c. Code your figure

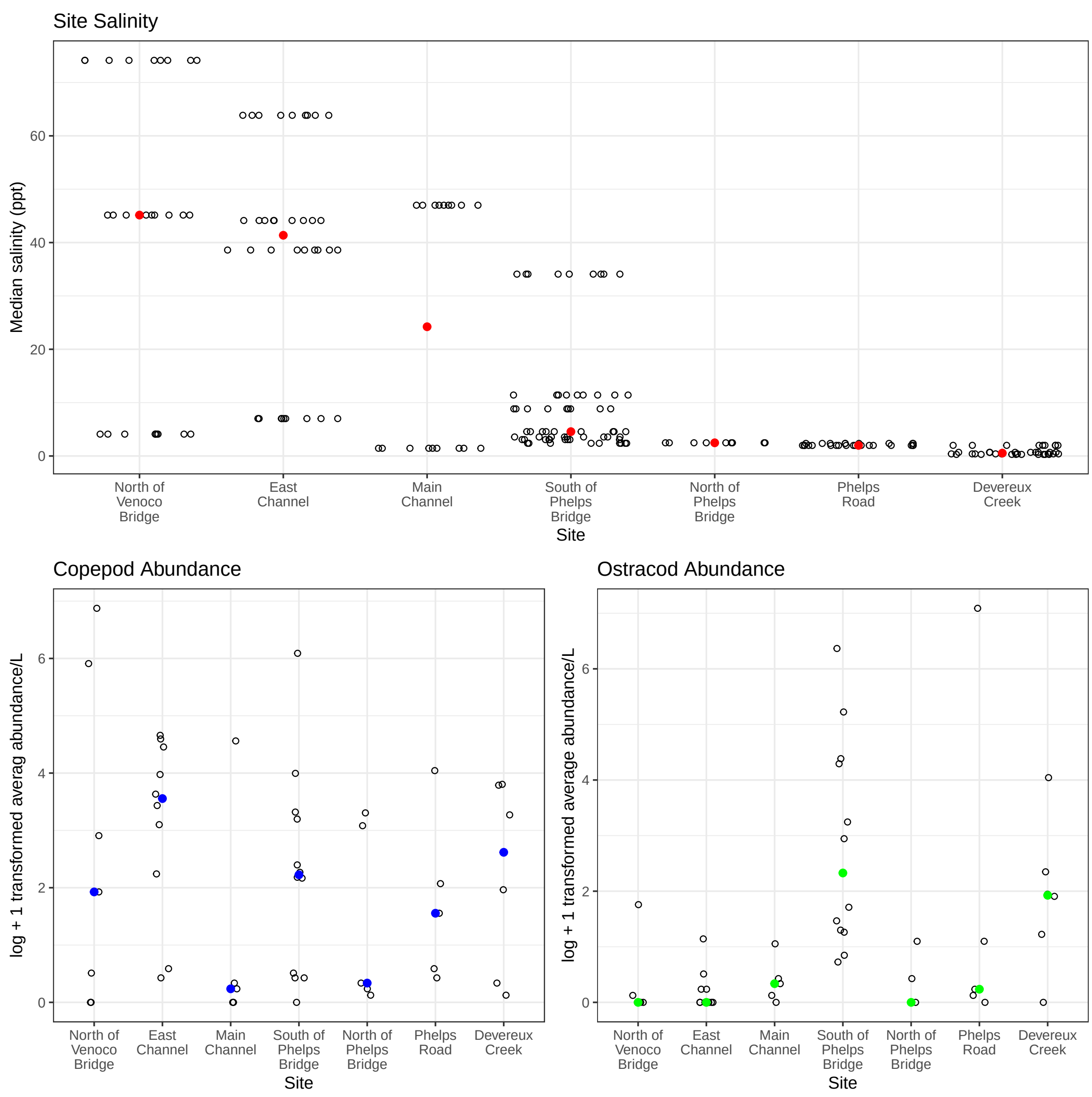
```
#  base layer: create an object that stores the plot
ostracod_plot <- ggplot(data = ostracod,
                        # x = site
                        # y = log + 1 transformed average abundance/L
```

```
                          mapping = aes(x = site_full,
                                        y = ostracod_log_avg_per_lit)) +

  # first layer: geometry jitter points to represent each sample
  geom_jitter(height = 0,
              width = 0.1,
              shape = 21) +
  # points to represent median abundance
  stat_summary(geom = "point",
               fun = median,
               color = "green",
               size = 2) +
  # wrapping x axis labels
  scale_x_discrete(labels = label_wrap(width = 10)) +

  # labeling axes and title
  labs(title = "Ostracod Abundance",
       x = "Site",
       y = "log + 1 transformed average abundance/L") +

  # changed theme
  theme_bw()

# display plot
ostracod_plot
```



d. Create a multipanel figure

```
salinity_plot / (copepod_plot | ostracod_plot)
```



e. Interpret your figure

What are the differences between sites in copepod and ostracod abundance?

- Average copepod abundance/L was highest at East Channel and Devereux Creek at respective transformed values of around 4 and 3, and moderate at a value of around 2 at North of Venoco Bridge, South of Phelps Bridge, and Phelps Road. The highest average ostracod abundance/L was moderate around a transformed value of 2 at South of Phelps Bridge and also Devereux Creek, and low at pretty much every other site.

How does ostracod abundance relate (or not relate) to site-level differences in salinity?

- Ostracod abundance does not correlate to salinity site-level differences in salinity because the plots do not show the same type of trend for the salinity plot and for the ostracod abundance plot.

2. Visualizing total abundance as a function of salinity

a. Plan your figure

- x-axis: salinity (ppt)
- y-axis: log-transformed average abundance/L
- color: taxon
- geometry: geom_point()
- facets: by each taxon in aq_ins_clean with free scales

b. Wrangle your data

```
# change existing aq_in_clean object
aq_ins_clean <- aq_ins_clean |>

  # change taxon values to title case
   mutate(taxon = str_replace_all(taxon, "_", " ") |> str_to_title()) |>

  # group by the sample date and by taxon
  group_by(date, taxon) |>

  # add in a new column that is the sum of avg_per_lit per taxon per date
  mutate(total_avg_per_lit = sum(avg_per_lit, na.rm = TRUE)) |>

  # ungroup
  ungroup() |>
```

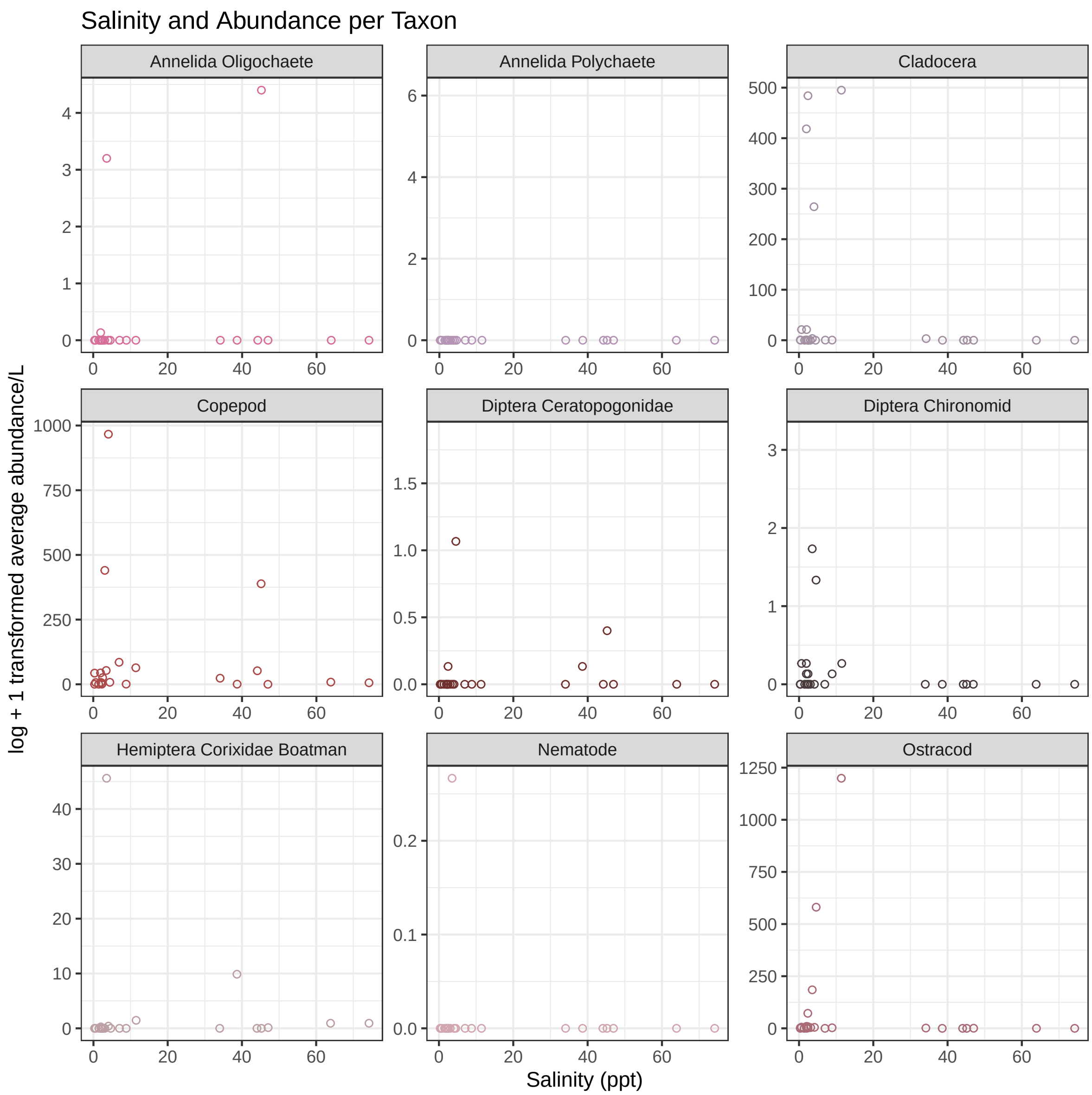
```r
  # keep only date, taxon, med_sal, and total avg/L
  select(date, taxon, med_sal, total_avg_per_lit) |>

  # reorder by max abundance
  arrange(desc(total_avg_per_lit))


# display 10 random rows
slice_sample(aq_ins_clean, n = 10)
```

```
# A tibble: 10 x 4
   date                taxon                    med_sal total_avg_per_lit
   <dttm>              <chr>                      <dbl>             <dbl>
 1 2022-02-23 00:00:00 Cladocera                 45.2               0.133
 2 2024-01-28 00:00:00 Annelida Polychaete        3.08              0
 3 2021-04-15 00:00:00 Ostracod                   3.57            185.
 4 2020-02-21 00:00:00 Diptera Chironomid         4.1               0
 5 2022-05-11 00:00:00 Nematode                   0.398             0
 6 2021-11-23 00:00:00 Annelida Oligochaete       1.99              0.133
 7 2022-05-05 00:00:00 Ostracod                  63.8               0.267
 8 2022-05-13 00:00:00 Diptera Ceratopogonidae   74.2               0
 9 2022-08-15 00:00:00 Copepod                   NA                 0
10 2023-11-11 00:00:00 Copepod                   38.6               0.533
```

c. Code your figure

```r
#  base layer: create an object that stores the plot
salinity_plot <- ggplot(data = aq_ins_clean,
                        # x = salinity
                        # y = log + 1 transformed average abundance/L
                        mapping = aes(x = med_sal,
                                      y = total_avg_per_lit,
                                      color = taxon)) +

  # first layer: geometry jitter points to represent each sample

  geom_jitter(height = 0,
              width = 0.1,
              shape = 21) +
```

```
  facet_wrap( ~taxon,
              scales = "free") +

  scale_color_manual(values = natparks.pals("LakeNakuru", n = 9)) +

  # labeling axes and title
  labs(title = "Salinity and Abundance per Taxon",
       x = "Salinity (ppt)",
       y = "log + 1 transformed average abundance/L") +

  # changed theme
  theme_bw() +

  # remove legend
  theme(legend.position = "none")


# display plot
salinity_plot
```



d. Interpret your figure